

Practical 4 : Digital Signature

AIM : Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CODE :

```
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes
from cryptography.exceptions import InvalidSignature
```

```
private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)
```

```
public_key = private_key.public_key()
```

```
message = input("Enter message: ").encode()
```

```
signature = private_key.sign(
    message,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)
```

```
print("\nSignature:", signature.hex())
```

```
try:
    public_key.verify(
        signature,
        message,
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("Signature verified successfully.")
```

except InvalidSignature:

```
print("Signature verification failed.")
```

```
modified_message = input("\nEnter message to verify: ").encode()
```

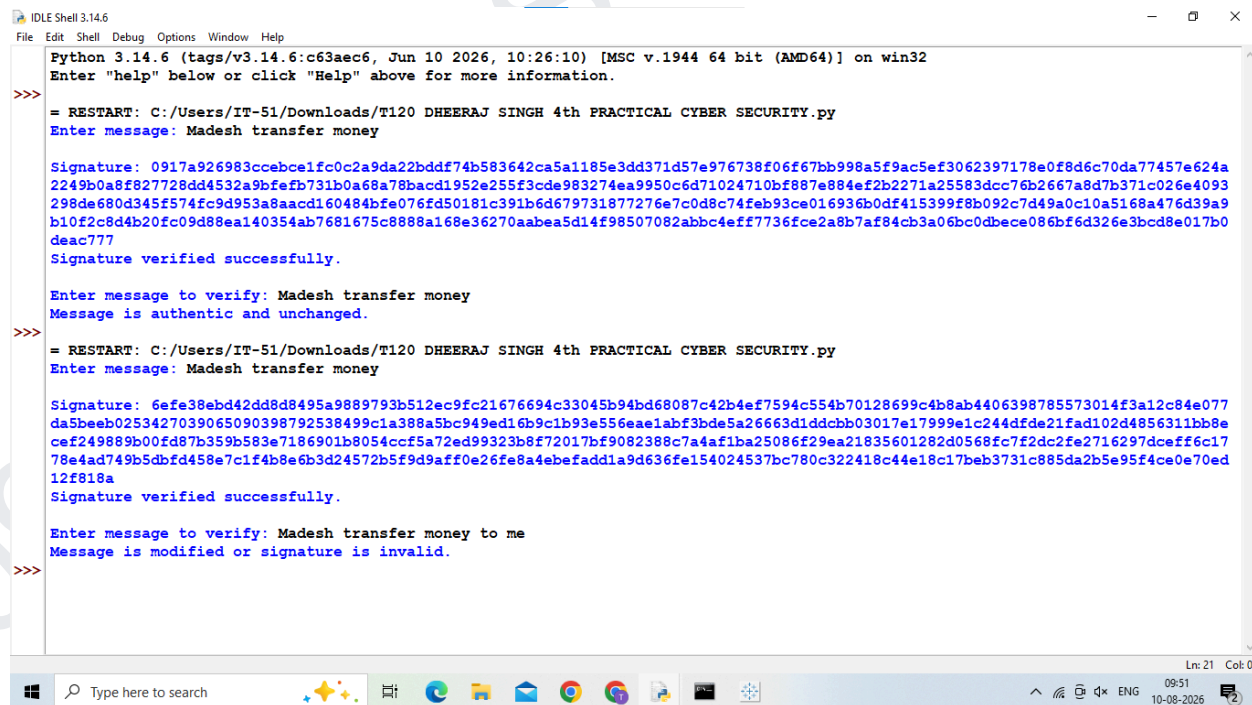
try:

```
public_key.verify(
    signature,
    modified_message,
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)
print("Message is authentic and unchanged.")
```

except InvalidSignature:

```
print("Message is modified or signature is invalid.")
```

OUTPUT :



```
IDLE Shell 3.14.6
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/IT-51/Downloads/T120 DHEERAJ SINGH 4th PRACTICAL CYBER SECURITY.py
Enter message: Madesh transfer money

Signature: 0917a926983ccebcbefc0c2a9da22bddf74b583642ca5a1185e3dd371d57e976738f06f67bb998a5f9ac5ef3062397178e0f8d6c70da77457e624a
2249b0a8f827728dd4532a9bfebf731b0a68a78bacd1952e255f3cde983274ea9950c6d71024710bf887e884ef2b2271a25583dcc76b2667a8d7b371c026e4093
298de680d345f574fc9d953a8aacd160484bfe076fd50181c391b6d679731877276e7c0d8c74feb93ce016936b0df415399f8b092c7d49a0c10a5168a476d39a9
b10f2c8d4b20fc09d88ea140354ab7681675c8888a168e36270aabea5d14f98507082abbc4eff7736fce2a8b7af84cb3a06bc0dbece086bf6d326e3bcd8e017b0
deac777
Signature verified successfully.

Enter message to verify: Madesh transfer money
Message is authentic and unchanged.

>>> = RESTART: C:/Users/IT-51/Downloads/T120 DHEERAJ SINGH 4th PRACTICAL CYBER SECURITY.py
Enter message: Madesh transfer money

Signature: 6efe38ebd42dd8d8495a9889793b512ec9fc21676694c33045b94bd68087c42b4ef7594c554b70128699c4b8ab4406398785573014f3a12c84e077
da5beeb0253427039065090398792538499c1a388a5bc949ed16b9c1b93e556eae1abf3bde5a26663d1ddcb03017e17999e1c244dfde21fad102d4856311bb8e
cef249889b00fd87b359b583e7186901b8054ccf5a72ed99323b8f72017bf9082388c7a4af1ba25086f29ea21835601282d0568fc7f2dc2fe2716297dceff6c17
78e4ad749b5dbfd458e7c1f4b8e6b3d24572b5f9d9aff0e26fe8a4ebefadd1a9d636fe154024537bc780c322418c44e18c17beb3731c885da2b5e95f4ce0e70ed
12f818a
Signature verified successfully.

Enter message to verify: Madesh transfer money to me
Message is modified or signature is invalid.

>>>
```